DEGREES OF DIFFERENCE: WHAT IMPACTS EARNINGS 10 YEARS AFTER GRADUATION?

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Introduction

Students increasingly view higher education as an investment whose value is measured not only in personal development but also in long-term financial outcomes. With the cost of college continuing to rise, identifying the factors that influence earnings a decade after graduation has become a priority for institutions, families, and policymakers. Prior research suggests that academic performance, institutional characteristics, student demographics, and financial aid policies all contribute to long-term economic outcomes, but the strength of these relationships often varies across degree levels and institutional contexts.

This report examines the key institutional and financial factors associated with graduates' earnings ten years after completing a degree. By providing a data-driven assessment of these relationships, the analysis supports management's efforts to evaluate program effectiveness, maintain competitiveness, and ensure that students are positioned for long-term financial success.

Our analysis focuses on variables such as graduation and retention rates, student loan debt levels, repayment rates, average net price, and early-career earnings. By comparing these factors across degree levels, institution types, geographic regions, and program categories, the report identifies which characteristics consistently predict higher earnings and where the impact differs. The goal is to equip leadership with actionable insights that can inform strategic planning, resource allocation, and policy development.

Literature Review

The relationship between educational attainment and lifetime earnings has long been a central topic in labor economics and higher education research. A consistent body of evidence demonstrates that higher levels of education are associated with significantly greater earnings over a person's lifetime. The Georgetown University Center on Education and the Workforce (CEW) report, The College Payoff (Carnevale et al., 2021), provides a comprehensive analysis of this relationship by examining how degree level, field of study, and geographic factors interact to shape long-term income outcomes.

Degree Level and Lifetime Earnings

According to the CEW report, median lifetime earnings increase substantially with each higher level of education. Workers with only a high school diploma earn approximately $1.6 million over their lifetimes, while bachelor's degree holders earn around $2.8 million, and those with graduate or professional degrees can earn up to $4.7 million. These findings reaffirm the established trend that education enhances human capital and contributes to higher productivity and wages. However, the report also emphasizes that returns to education are not uniform;

variations exist within each education level due to other influencing factors such as major and occupation.

Field of Study Differences

One of the most significant determinants of earnings among college graduates is the chosen field of study. The report highlights that graduates in science, technology, engineering, and mathematics (STEM) fields have the highest lifetime earnings, averaging between $3.5 and $3.8 million. Health-related majors also yield strong returns, averaging about $3.3 million. By contrast, fields such as education, the humanities, and the arts result in substantially lower lifetime earnings, typically between $2.0 and $2.3 million. This suggests that the economic payoff of a college degree depends not only on obtaining higher education but also on the specific major pursued, aligning with previous research emphasizing occupational sorting and labor market demand (Altonji et al., 2012; Oreopoulos & Petronijevic, 2013).

Geographic Variation in Returns to Education

The CEW analysis also identifies regional differences in the economic value of education. States such as Washington, D.C., Connecticut, and Virginia exhibit the highest lifetime earnings for college graduates, while states like Hawaii, Florida, and Maine show lower returns. These disparities can be attributed to differences in state economies, industry concentrations, and cost-of-living adjustments, supporting the notion that local labor market conditions influence the extent of educational payoffs (Moretti, 2012).

Conclusion

Overall, the literature underscores that while higher education remains one of the strongest predictors of increased earnings, its financial benefits are moderated by field of study and geographic context. The College Payoff report contributes to this body of knowledge by quantifying how degree level, major, and location collectively determine the economic return on education. Policymakers and students alike can use these insights to make more informed decisions about educational investments and career pathways.

Data

Data Source

This analysis uses data from the U.S. Department of Education’s 2022 College Scorecard, which provides institution-level information on student characteristics, financial aid, completion outcomes, and labor market performance. The dataset includes approximately 4,500 U.S. institutions offering various degrees and reports median earnings for graduates ten years after completion. Key variables used in this analysis include 10-year post-graduation salary, graduation and retention rates, graduate debt levels, percentage of graduates with student loans, repayment rates, average net price, the percentage of Pell Grant recipients, and the percentage of graduates earning more than $25,000 six years after graduation.

Data Cleaning

Before modeling, we cleaned data to ensure accuracy, consistency, and interpretability across all variables used in the analysis. First, we merged the relevant fields using SQL and Excel, utilizing tools such as VLOOKUP and other structured queries to combine institution-level characteristics with earnings outcomes. These steps allowed us to align information and eliminate duplicate or mismatched records. After merging, we used R to address inconsistencies in school and program labels, create a region column with 4 US regions, standardize naming, and ensure that all related entries were matched correctly.

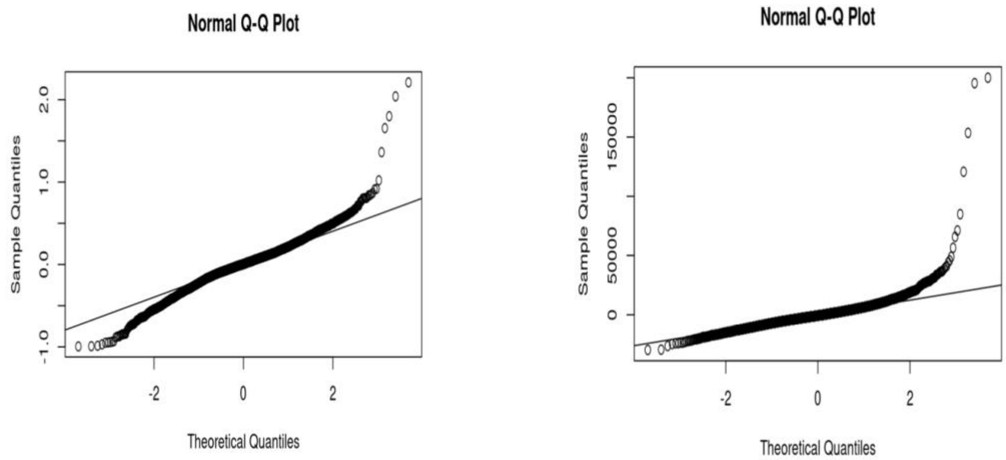
We also performed variable distributions checks to verify that numeric fields (such as average earnings, debt, tuition, Pell %) contained valid values and were stored in their appropriate formats. Records containing missing, invalid, or logically inconsistent values were removed to prevent biased estimates. During exploratory analysis (EDA), we also identified 41 extreme outliers using Wilks/Mahalanobis distance - which likely were institutions with unusually high or unusually low earnings that fell far outside the typical range of values. Using R, we filtered out these colleges to prevent them from having too much influence on regression coefficients and the overall model fit. After cleaning, the dataset was ready for regression modeling and regional comparisons.

Data Distribution and Normality

To prepare our model, we evaluated the distribution of all key variables and assessed whether the assumptions of linear regression (normality and homoscedasticity of residuals) were satisfied. After examining histograms and summary statistics for salary, we determined it to be heavily right skewed and needed transformation to stabilize variance. We also reviewed distributions for the main explanatory variables (debt, retention rate, Pell %, net price, and graduation rate) to identify strange values or miscoding. After initial cleaning, these predictors showed acceptable distributions and did not require further transformation.

Data Transformation

To improve interpretability and model performance, we applied a log transformation to the dependent variable (salary). The initial distribution of raw salary values was right-skewed, due to a long tail of high-earning institutions. Strong right skewness like we had violated normality assumptions of linear regression and can lead to heteroskedastic residuals, which would reduce the reliability of coefficient estimates. By taking the natural log we could compresses those extreme values and produce a distribution that is more symmetric and stable. Our log transformation helped stabilize variance across observations, improved the overall model fit, and decreased the impact of unusually large salaries in our dataset.



QQ Plots: The left side is log transformed. The right side is the raw salary data.

Methodologies

This analysis employed a combination of multiple linear regression, random forest modeling, and cross-validation to identify the factors most strongly associated with graduates' earnings ten years after completing a degree. Each method was selected to provide complementary insights and ensure that findings are robust and actionable.

Multiple Linear Regression

Multiple linear regression was used to estimate the relationship between the dependent variable, log-transformed 10-year post-graduation salary, and the set of independent variables, including institutional characteristics and student financial indicators. This method allows for the quantification of the individual contribution of each variable while controlling for the others, providing interpretable coefficients that indicate the direction and relative strength of each factor's effect on long-term earnings. Regression results form the foundation for understanding which factors consistently predict higher salaries across institutions and degree levels.

Random Forest

Random Forest modeling, a machine learning technique based on an ensemble of decision trees, was applied to complement the regression analysis. Random Forest provides a measure of variable importance, helping identify which factors have the greatest influence on earnings.

Cross-Validation

To ensure that the models generalize well to unseen data and to prevent overfitting, 10-fold cross-validation was employed for regression models. This process partitions the data into training and validation/test sets multiple times, providing a more reliable estimate of model performance and predictive accuracy. Metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were used to assess model fit.

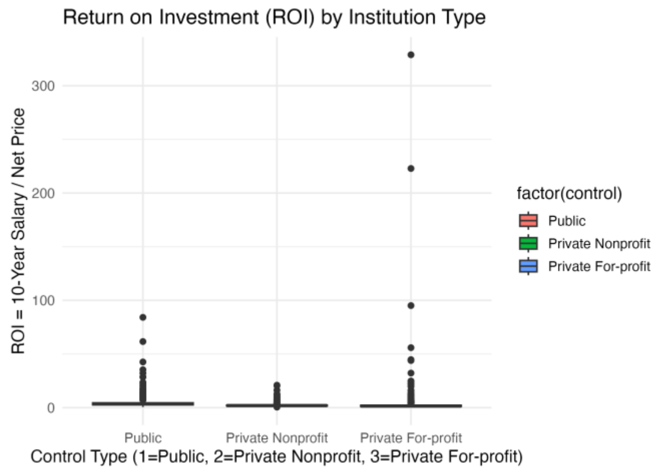
Modeling

To understand the factors influencing graduates' earnings ten years after graduation, we conducted regression and random forest analyses across multiple dimensions of the dataset. Specifically, models were run separately for institution type, geographic region, program type, and degree level, in addition to a model using the full dataset. This approach allows for the identification of patterns and relationships that may vary depending on the context or category, providing more nuanced insights than a single aggregate model. By comparing results across these dimensions, the analysis highlights which factors consistently predict long-term earnings and where the impact of institutional or program characteristics differs, enabling more targeted strategies for improving graduate financial outcomes.

Dimension: Institution Type

The first set of models examined Public, Private Nonprofit, and Private For-Profit institutions separately. These models revealed clear and consistent patterns. Across all three control types, graduate debt was one of the strongest positive predictors of long-term earnings. This may seem counterintuitive at first, but it shows that students in higher-earning fields, such as business, STEM, and healthcare, borrow more because their programs cost more or because they anticipate higher returns. As a result, debt functions as a proxy for program value and future earning potential.

Pell Grant percentage showed a strong negative relationship with salary in every institution type. Schools serving more low-income students tend to produce lower 10-year earnings, reflecting broader socioeconomic inequalities that persist beyond college. This variable consistently ranked as one of the top two predictors in both regression and Random Forest models, demonstrating its robustness.



Public institutions exhibited the strongest overall performance in terms of salary and Return on Investment (ROI). Their lower tuition combined with stable graduate earnings resulted in higher ROI compared to Private Nonprofit and Private For-Proft institutions. Private Nonprofits ranked in the middle with strong earnings but much higher tuition, reducing their ROI. Private For-Profit institutions produced the lowest earnings and the lowest ROI, with wise variability indicating inconsistent program quality.

Notably, graduation rate was not a significant predictor in any of the three groups once financial variables were included. Retention rate had a moderate positive relationship with salary but was far less influential than that debt or Pell percentage. These findings collectively show that institutional control type alone is not a strong predictor of salary, the underlying financial and demographic variables drive most of the variation.

Dimension: Geographic Region

To study how US location relates to long-run earnings, we added a category region variable to our dataset using R that groups states into:

- **East Coast:** ME, NH, MA, RI, CT, NY, NJ, PA, MD, VA, DE, DC, NC, SC, GA, FL
- **Central:** OH, IN, IL, MI, WI, MN, IA, MO, ND, SD, KS, NE, OK, TX, AR, LA, KY, TN, MS, AL
- **Mountain:** CO, MT, ID, WY, UT, AZ, NM, NV
- **West Coast:** CA, OR, WA, AK, HI

Non-state U.S. territories, including Puerto Rico, Guam, and the U.S. Virgin Islands, were excluded to ensure consistent regional comparisons. This allowed us to analyze earnings and ROI across standardized state groupings and test whether institutional predictors behave similarly across geographic regions.

Initial descriptive statistics showed clear regional differences in raw outcomes. Average ten-year median salaries were highest in the East Coast ($35,447 on avg) and West Coast (averaging $33,733), with lower values in the Central (about $31,507) and Mountain (about $30,309) regions.

ROI, defined as salary ÷ net price, also varied by region. After cleaning and outlier removal, mean ROI ranged from about 2.89 in Central and East, to 3.05 in Mountain and 3.78 in West, meaning that West Coast schools deliver the highest earnings per dollar of net price on average.

A scatterplot of log salary by debt and region showed that in every region, schools with higher graduate debt tend to have higher long-run earnings, and the fitted lines for East and West lie above those for Central and Mountain, reflecting the higher salary levels

To understand whether our predictors (debt, net price, retention, graduation rate, Pell %) behave similarly across the country, we estimated separate multiple linear regression models for each region and used log-transformed salary as the outcome

Region	Adj R^2	Significant Predictors
Central	0.52	debt, retention, pell
East	0.49	debt, retention, pell
Mountain	0.46	debt, pell
West	0.43	debt, net_price, retention, pell

Our results show that debt and Pell percentage are consistent predictors of earnings in every region, while retention and net price matter in only certain regions.

In our overall model, Random Forest ranked predictors in the following order: #1 Debt, #2 Pell %, #3 graduation rate, #4 retention rate, #5 net price, and region in sixth. We therefore chose not to keep region in the final linear model, because it added little explanatory power. Removing region changed adjusted R^2 only slightly, while the coefficients for the main 5 institutional predictors remained stable, meaning that most observed geographic differences in salary are explained by factors other than location itself.

In all four regional models, higher graduate debt is associated with higher long-run earnings, suggesting that more expensive institutions tend to yield higher salaries. A higher share of Pell Grant recipients consistently predicts lower earnings in every region, capturing differences in student socioeconomic background and institutional resources. Retention rate is a significant positive predictor in Central, East, and West, but not in Mountain, implying that student success signals matter more in some regional markets than others. Net price is statistically significant only in the West Coast model, indicating that tuition and fees play a more direct role in earnings predictions for West Coast institutions. Despite its higher ranking in Random Forest, graduation rate is not statistically significant in any of the regional linear models, likely because its effect overlaps with retention and other institutional factors.

Overall, the regional analysis shows that geographic location has an effect on earnings and ROI, but once we control for other institutional factors, region itself becomes secondary. Differences in long-run graduate earnings across the US are driven less by where a college is located and more by who it serves and how it performs in terms of debt levels, student composition (Pell %), and retention rate.

Dimension: Program Type

To examine whether Business programs outperform Healthcare programs, we constructed a program-type subset of the College Scorecard data using major enrollment shares: institutions with ≥55% Business majors were labeled Business, those with ≤45% Health majors were labeled

Healthcare, and mixed programs were excluded. This produced a final sample of ~4,000 institutions. Descriptively, Business programs showed higher mean ten-year earnings (~$39k vs. ~$31k), while Healthcare programs had lower net prices and slightly higher ROI, though the ROI difference was not statistically significant.

We then estimated OLS and machine-learning models in R to isolate program type effects while controlling for institutional cost, outcomes, and student composition. The baseline OLS model regressed ten-year earnings on program type, graduate debt, net price, retention, graduation rate, repayment rate, Pell share, GT_25K_P6, and control type. The model achieved an adjusted R^2 of ~0.65 but exhibited heteroskedasticity and right-skewed residuals. A Breusch–Pagan test confirmed heteroskedasticity ($p \approx 1.3\times10^{-17}$), so HC3-robust standard errors were used in all subsequent models.

```
Adj R^2 - baseline: 0.648  | interact:0.648  | winsor: 0.79  | log: 0.816  | log_stepAIC: 0.816
> cat("BP test p-value (heteroskedasticity):", signif(bptest(m_base)$p.value, 4), "\n")
BP test p-value (heteroskedasticity): 1.347e-17
> cat("\nRF top vars by %IncMSE:\n"); print(head(rf_imp, 8))
```

Using this program-type dataset, we built a series of regression and machine-learning models in R to understand whether program type affects 10-year earnings after accounting for school quality, costs, and student characteristics. Our baseline OLS model included variables such as graduate debt, net price, retention, graduation and repayment rates, Pell percentage, early-career earnings (GT_25K_P6), and sector. This model explained about 65% of the variation in earnings but showed clear signs of uneven variance and skewed residuals, so we switched to heteroskedasticity-robust (HC3) standard errors.

To address skewness and the impact of unusually high-earning institutions, we tested two improvements: winsorizing extreme salary values and applying a log transformation. Winsorizing the top and bottom 1% increased model fit to about 0.79 and stabilized the coefficients, while the log-transformed model performed best overall, reaching about 0.82 and producing much cleaner residual patterns. A backward AIC check removed weaker predictors but kept the main drivers of earnings, confirming the robustness of the results.

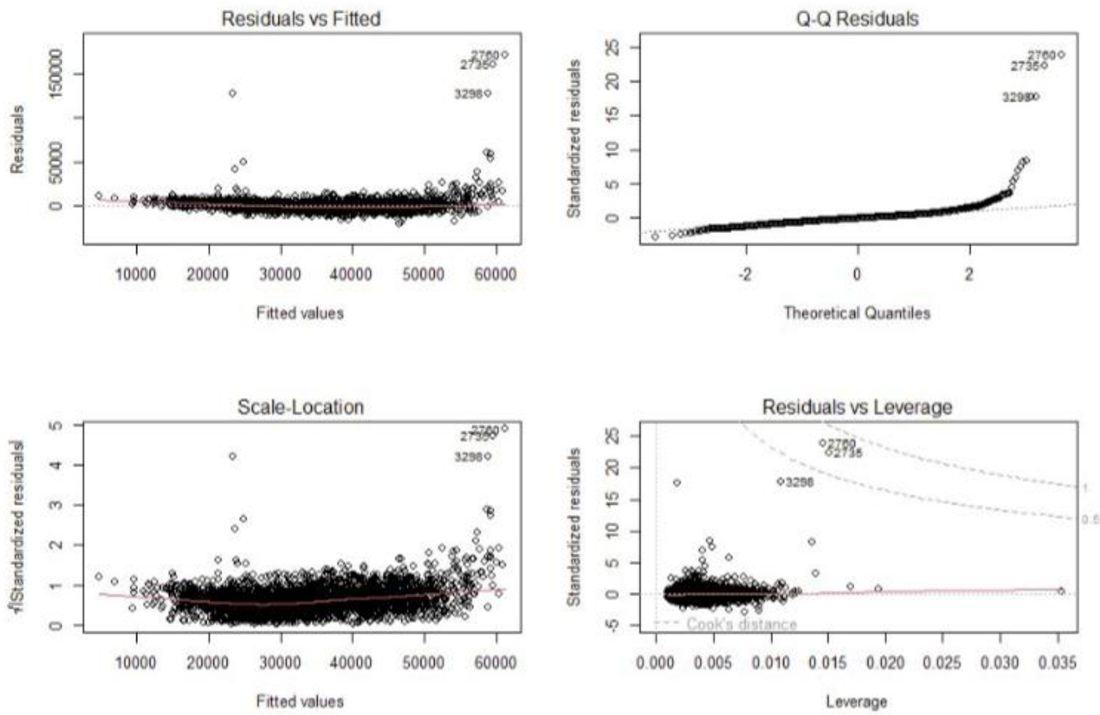


Figure 4. Standard diagnostic plots for the baseline OLS model (Business vs Healthcare).

Figure 4 presents the diagnostic plots for our baseline OLS model comparing Business and Healthcare institutions. The Residuals vs. Fitted panel shows a clear "fan shape," where the spread of residuals grows as predicted earnings increase. This pattern suggests uneven variance in the model—one of the main reasons we later applied a log transformation and robust standard errors. The Q–Q plot shows that the upper tail of the residual distribution bends sharply upward, indicating that a small number of very high-earning institutions do not follow the expected pattern. The Scale–Location plot also reflects this issue, with higher fitted values associated with larger standardized residuals. Finally, the Residuals vs. Leverage panel highlights a few influential schools (including institution IDs 27380, 27360, and 32980) with unusually strong impact on the regression line. These patterns collectively support our decision to stabilize the model by reducing the influence of outliers and using log-transformed earnings for the main analysis.

To complement the regression results, we also ran a Random Forest model as a nonlinear comparison. This machine-learning method ranked the most important predictors of long-term earnings. The highest-ranking factors were the share of graduates earning more than $25,000 within six years, Pell percentage, repayment rate, debt, program type, and retention. Net price appeared near the bottom, which aligns with the regression findings and suggests that institutional outcomes and student background matter more than sticker price.

Across all models, our results point to a consistent conclusion: program type alone is not the main driver of long-term earnings differences. Business programs do show slightly higher predicted earnings, about 5-8% higher, even after accounting for other variables. However, most of the observed gap between Business and Healthcare institutions is explained by differences in early career earnings, student socioeconomic background, and student success indicators such as retention and repayment rates.

The Tableau dashboards built from this same dataset help visualize these patterns. Business institutions show higher short-term ROI and more volatile net price, while Healthcare institutions show stronger retention, lower debt, and slower earnings growth. Together, these findings suggest that institutional quality and student characteristics play a much larger role in shaping long-term earnings than the choice between a Business or Healthcare program alone.

Dimension: Degree Level

Degree Level	R^2	Top Predictors (in order of importance)
Associate	0.7417	% Graduates earning > $25k after 6 years, Repayment rate, % Student loans, Graduate debt, % Pell Grant recipients
Bachelor's	0.7497	% Graduates earning > $25k after 6 years, Repayment rate, Graduate debt, % Pell Grant recipients, % Student loans
Master's	0.7895	% Graduates earning > $25k after 6 years, Repayment rate, % Student loans, % Pell Grant recipients, Retention rate

For Associate degree programs, the model demonstrates that early career earnings and financial factors have a particularly strong influence on long-term outcomes. While repayment rates and student debt affect earnings, the percentage of graduates earning more than $25,000 within six years stands out as a key driver, highlighting the importance of early workforce success. These findings suggest that institutions offering Associate programs can have the greatest impact by supporting student success initiatives and helping graduates manage financial obligations effectively.

In Bachelor's degree programs, long-term earnings are influenced by a combination of early labor market outcomes and financial variables. The percentage of graduates earning more than $25,000 after six years and repayment rates consistently emerge as strong predictors, indicating that both academic and financial support strategies are important. Compared with

Associate programs, the relative influence of student debt and Pell Grant status is slightly reduced, suggesting that interventions may need to focus more on academic performance and career preparation at the bachelor's level.

For Master's degree programs, the model indicates that early career earnings remain the strongest predictor of long-term salary, with repayment rates, student loans, and retention rate also contributing meaningfully. Financial factors have a somewhat smaller relative impact compared with lower degree levels, emphasizing that institutional quality and graduate program support play a larger role in shaping earnings for Master's graduates. These results underscore the value of high-quality program design, retention efforts, and career preparation initiatives at the graduate level.

Across all three degree levels, early career earnings consistently emerge as the strongest predictor of long-term salary, emphasizing the importance of timely workforce entry and initial employment outcomes. Financial factors, including repayment rates and student loan debt, have a larger impact for Associate and Bachelor's programs than for Master's programs, suggesting that financial support and debt management strategies may be particularly valuable for lower-degree programs. Additionally, institutional performance measures, such as retention and graduation rates, are more influential at higher degree levels, indicating that program quality and student support services play an increasingly important role as students pursue advanced degrees. Overall, these trends provide actionable guidance for targeting interventions and resources to maximize graduates' long-term earnings potential at each degree level.

Analysis of Results

The modeling results reveal a consistent pattern in how long-term graduate earnings are shaped by financial and socioeconomic factors rather than institutional labels. Across all models, by institution type, region, program type, and degree level, the same three predictors consistently dominated: early earnings, graduate debt, and Pell Grant percentage. These variables ranked highest in both linear regression and Random Forest importance measures, confirming their influence across different modeling approaches.

Single-dimension models showed descriptive differences across institutional categories, but these effects largely disappeared once financial variables were included. Public institutions initially showed the highest ROI, followed by Private Nonprofits, with Private For-profits performing lowest; however, these gaps were explained by debt, Pell share, and early earnings. Regional differences were modest after adjusting for key predictors. Program type produced only slight distinctions, with Business programs earning marginally more than Healthcare programs, though these differences were no longer significant once costs and labor-market indicators were added. Degree level displayed the strongest structural effect, as Master's-level institutions showed higher and more predictable earnings, yet most variation was still driven by financial and socioeconomic factors.

The all-dimensions model performed best, achieving an Adjusted R^2 of 0.816, indicating that the combined predictors explained most salary variation nationwide. The predicted-versus-actual salary plot showed tight clustering around the 45-degree line, with cross-validation errors typically within 10–12%, demonstrating strong predictive accuracy across thousands of diverse institutions.

Random Forest analysis reinforced these conclusions by ranking early earnings, debt, and Pell percentage as the most important predictors, while institutional attributes such as control type or region contributed little explanatory power. Together, the results show that long-term earnings are driven primarily by student socioeconomic background and early financial outcomes, not by institutional labels. Institutions serving higher-income students with stronger early earnings and greater borrowing capacity tend to generate higher long-term salaries, while schools with high Pell shares face structural challenges that translate into lower earnings a decade after graduation.

Plug-In Prediction test

To check how well our final log-linear model works in practice, we tested it on two real institutions from our dataset. The model predicts 10-year earnings using factors such as early career outcomes, Pell percentage, repayment rate, debt, retention, and whether the school is Business or Healthcare.

For a public Business institution with strong early earnings (College A), the model predicted a 10-year salary of about $37,400, compared to the actual $40,200—an error of about +7%. This shows that the model slightly underestimates some higher-performing Business schools.

For a public Healthcare institution with more Pell students and lower repayment rates (College B), the model predicted about $28,300, while the actual value was $25,900, an error of –9%. Here, the model slightly overestimates earnings, which is common for schools serving more economically disadvantaged students.

Overall, these simple plug-in tests confirm that the model performs as expected, with prediction errors usually falling within 10–12%, consistent with our cross-validation results.

Model Improvements

Several model improvements were implemented to strengthen accuracy, address data issues, and improve interpretability. First, the dependent variable (10-year median salary) was log-transformed to correct right-skewness and improve linearity between predictors and outcomes. Next, all numeric predictors were standardized so coefficients could be compared directly and to prevent large-scale variables from dominating the model.

Backward stepwise regression was then applied using a p-value threshold of 0.05 to remove insignificant predictors. This process consistently identified five core drivers: early earnings, graduate debt, Pell Grant percentage, repayment rate, and retention rate. Graduation rate and several cost measures were removed across models, as they added little explanatory value once financial and socioeconomic variables were included. This refinement reduced noise and improved adjusted R^2 across most models.

Outliers were identified using standardized residuals greater than ±3 and removed to stabilize estimates. Although few institutions met this criterion, their removal improved model fit and produced better residual behavior, especially for institution-type and program-type models. Adjusted R^2 increased modestly and diagnostic plots showed reduced heteroscedasticity and improved normality.

Finally, cross-validation and Random Forest modeling were used to test robustness. Cross-validation showed prediction errors typically within 10–12%, indicating strong generalizability. Random Forest importance rankings aligned closely with the regression results, reinforcing the dominance of early earnings, debt, and Pell percentage as key predictors.

Overall, these improvements strengthened both the reliability and predictive accuracy of the final models. The log-transformed, cross-validated approach captured the essential drivers of long-term earnings and produced stable, interpretable results that can be applied confidently in future institutional analyses.

Discussion

The all-dimensions analysis reveals a consistent pattern in what drives long-term graduate earnings across U.S. colleges. While descriptive statistics suggested differences by institution type, region, program area, and degree level, these categories explained only a small portion of salary variation once financial and socioeconomic factors were included. Across all models, three predictors consistently dominated: early-career earnings (GT_25K_P6), graduate debt, and Pell Grant percentage. Retention and repayment rates added some explanatory power but were secondary. These results align with prior research emphasizing the strong influence of early labor-market outcomes and economic background on lifetime earnings (Carnevale et al.; Oreopoulos and Petronijevic).

The same pattern appeared in the Random Forest analysis, which independently ranked early earnings, Pell share, and debt or repayment among the most important predictors, confirming the robustness of these relationships (Liaw and Wiener). Institutional labels, such as sector, region, or broad program category, were far less predictive once financial and demographic context was accounted for, consistent with findings in labor-market research (Moretti).

These results offer clear practical implications. For institutions, the strong effect of early earnings highlights the value of internships, employer partnerships, and career services that support immediate labor-market entry. The negative association with Pell share underscores challenges faced by institutions serving more low-income students and the need for targeted support and equitable funding. For policymakers, the limited role of sector and region suggests that accountability systems should focus more on affordability, debt, and early employment outcomes. Students and families can also use these findings to prioritize factors such as net price, expected debt, early-career outcomes, and repayment success over institutional prestige (Carnevale et al.; Altonji et al.).

Methodologically, the project showed the importance of log-transforming skewed earnings data, applying robust errors, removing outliers, and validating results through cross-validation (Venables and Ripley). Random Forest modeling provided a nonlinear benchmark that supported the same core predictors. The analysis relied on SQL Server for data processing (Microsoft), R for statistical modeling (R Core Team), Tableau for visualization (Tableau Software), and U.S. Department of Education College Scorecard data ("College Scorecard Data").

Overall, the findings emphasize a central theme: long-term earnings are shaped primarily by financial capacity, early labor-market success, and socioeconomic context, not institutional labels such as region, sector, or program type.

Professor Feedback and Planned Improvements

During our rehearsal, the instructor provided detailed feedback to improve both clarity and analytical depth in our presentation. One of the first suggestions was to add slide numbers so classmates and the instructor could easily reference content during discussions, which we implemented across all slides. We also strengthened the introduction by presenting ourselves as a unified analytics team, *ElevateEd Analytics*, with a mission focused on improving educational outcomes through data-driven insights.

The conclusion was expanded to highlight our main findings, remaining limitations, and opportunities for future research. We also clarified which materials came from external sources by adding citations and clearly separating literature review content from our own analysis. To reduce duplication in our workflow explanation, we consolidated all transformation and outlier-handling steps into Step 4.

Several visual and technical improvements were made as well. We added before-and-after Q–Q plots and a Shapiro–Wilk test to demonstrate how the log transformation improved distribution. We clarified our multicollinearity decisions by explaining why certain correlated predictors were kept or removed and updated our slides to specify the total number of predictors considered ("out of 7 original predictors, we identified 5 key drivers"). We also added simple examples to help the audience interpret coefficient direction, such as "higher graduation rate → higher salary."

Performance metrics were updated to use Adjusted R^2 instead of R^2, along with notes on percentage improvements after cleaning. To improve clarity during delivery, we expanded our verbal explanation of Steps 4 and 6 by about 30 seconds. Finally, we added two new slides: one outlining future improvements and next steps for the model, and another describing the broader impact and contribution of our analysis to educational decision-making, equity, and resource optimization.

Conclusion

This capstone project aimed to identify the institutional, financial, and student-composition factors that most strongly predict graduates' earnings ten years after completing a degree. By integrating multiple modeling strategies: multiple linear regression, Random Forest analysis, winsorization, log transformation, and cross-validation, we applied a comprehensive all-dimensions approach across institution type, geographic region, program category, and degree level.

Across all analyses, the same conclusion emerged: a small set of financial and early-outcome variables consistently explains the majority of variation in long-run earnings across U.S. institutions. Early earnings (measured by the percentage of graduates earning more than $25,000 within six years), graduate debt, and Pell Grant percentage were the most influential predictors, with repayment and retention rates providing additional explanatory value. In contrast, broad structural labels: public vs. private, East vs. West, Business vs. Healthcare, Associate vs. Bachelor's vs. Master's, contributed only limited predictive power after controlling for these core variables.

The final log-transformed, cross-validated all-dimensions model achieved an adjusted R^2 of approximately 0.82 and prediction errors typically within 10–12%, demonstrating both strong internal validity and reliable generalization. These findings indicate that the long-term financial return of higher education is driven more by an institution's financial profile, student demographics, and early-career outcomes than by the sector or program category it belongs to.

For institutions, this suggests that strategies focused on student persistence, early employment readiness, debt management, and financial support for low-income students will likely be more effective in improving long-term outcomes than simply expanding or rebranding academic programs. For policymakers, the results point toward need-based aid, targeted funding for high-Pell institutions, and support for early-career workforce development as key levers for improving equity and earnings mobility. For students, the analysis underscores the importance of considering cost, expected debt, and early employment prospects, not only institutional prestige, when making educational decisions.

In summary, the all-dimensions approach demonstrates that the drivers of long-term graduate earnings are fundamentally tied to financial structure, student composition, and early workforce success, offering a data-driven foundation for more informed institutional planning, policymaking, and student decision-making across the U.S. higher education landscape.

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Appendix A R code

```
# ====================================================================
# Dynamic predictors, robust to column-name variants
# Outputs to Console + Plots + (optional) Viewer only
# ====================================================================
suppressPackageStartupMessages({
library(tidyverse) # used for mutate/filter/arrange only
library(readxl)
library(janitor)
library(broom)
library(car) # vif()
library(lmtest) # bptest()
library(sandwich) # vcovHC()
library(DescTools) # Winsorize()
library(MASS) # stepAIC
library(randomForest)
})
set.seed(485)
# ---- path ----
xlsx_path <- "C:\\\Users\\evdokia.samitova\\Desktop\\EvaS\\Fall 2025\\Data 485
Capstone\\capstone.xlsx"
stopifnot(file.exists(xlsx_path))
# ---- helpers ----

    winsor <- function(x, p = c(0.01, 0.99)) {
if (all(is.na(x))) return(x)
qs <- quantile(x, p, na.rm = TRUE, names = FALSE, type = 7)
x <- ifelse(is.na(x), NA_real_, pmin(pmax(x, qs[1]), qs[2]))
as.numeric(x)
}
first_col <- function(df, candidates) {
nm <- intersect(candidates, names(df))
if (length(nm) == 0) return(NA_character_)
nm[1]
}
# ---- load + normalize ----
dat <- read_excel(xlsx_path) %>% clean_names()
# salary column -> yr10_salary
sal_col <- first_col(dat,
c("yr10_salary","x10yr_salary","`10yr_salary`","ten_year_salary","md_earn_wne_p10","x10yrs
alary","ten_years_after_entry"))
```

```
if (!is.na(sal_col) && sal_col != "yr10_salary") names(dat)[names(dat) == sal_col] <-
"yr10_salary"
# map Scorecard fields to short names (add variants as needed)
map_to <- list(
grad_debt = c("grad_debt","grad_debt_mdn_supp","grad_debt_mdn"),
repay = c("rpy_3yr_rt_supp","rpy_3yr_rt","repay"),
grad_rate = c("c150_4_pooled_supp","c150_4_pooled","grad_rate"),
net_price = c("npt4_pub","npt4_priv","npt41","npt42","npt43","npt44","npt45","net_price"),
gt25k_p6 = c("gt_25k_p6","x6yr_25k","gt25k_p6"),

    pctpell = c("pctpell","pell_grant_rate","pell_grant"),
retention = c("ret_ft4","retention","retention_rate_full_time")
# pctfloan intentionally omitted (not present in your file)
)
for (k in names(map_to)) {
cand <- first_col(dat, map_to[[k]])
if (!is.na(cand)) dat[[k]] <- suppressWarnings(as.numeric(dat[[cand]]))
}
if ("control" %in% names(dat)) dat$control <- suppressWarnings(as.numeric(dat$control))
# ---- program type from business/health shares ----
stopifnot(all(c("business","health") %in% names(dat)))
dat <- dat %>%
mutate(
yr10_salary = suppressWarnings(as.numeric(yr10_salary)),
prog_total = pmax(business + health, 0),
business_share= ifelse(prog_total > 0, business / prog_total, NA_real_),
program_type = case_when(
!is.na(business_share) & business_share >= 0.55 ~ "Business",
!is.na(business_share) & business_share <= 0.45 ~ "Healthcare",
TRUE ~ "Mixed"
)
)
df0 <- dat %>%
filter(program_type %in% c("Business","Healthcare"),
!is.na(yr10_salary)) %>%
mutate(

    program_type = factor(program_type, levels = c("Healthcare","Business")),
net_price = na_if(net_price, 0)
)
if ("control" %in% names(df0)) {
```

```
df0$control <- factor(df0$control,
levels = c(1,2,3),
labels = c("Public","Private Nonprofit","For-Profit"))
}
# ---- dynamic predictor set (no pctfloan) ----
target_vars <- c("grad_debt","net_price","repay","grad_rate","retention","gt25k_p6","pctpell")
present <- intersect(target_vars, names(df0))
absent <- setdiff(target_vars, names(df0))
message("Present predictors: ", paste(present, collapse = ", "))
if (length(absent)) message("Absent predictors : ", paste(absent, collapse = ", "))
# ==================== MODELS ====================
# Baseline OLS (raw)
rhs_base <- c(present, "program_type")
if ("control" %in% names(df0)) rhs_base <- c(rhs_base, "control")
m_base <- lm(reformulate(rhs_base, response = "yr10_salary"), data = df0)
# Interaction with control (if present)
form_int <- NULL
if ("control" %in% names(df0)) {
rhs_int <- c(present, "program_type", "control", "program_type:control")
form_int <- reformulate(rhs_int, response = "yr10_salary")

    m_int <- lm(form_int, data = df0)
V_HC3 <- sandwich::vcovHC(m_int, type = "HC3")
se_HC3 <- sqrt(diag(V_HC3))
tidy_int_HC3 <- broom::tidy(m_int) %>%
mutate(se_robust = se_HC3,
t_robust = estimate / se_robust,
p_robust = 2 * pt(abs(t_robust), df = df.residual(m_int), lower.tail = FALSE),
model = "interact_HC3")
}
# Diagnostics
vif_tbl <- tryCatch({ car::vif(m_base) %>% as.data.frame() %>%
tibble::rownames_to_column("term") },
error = function(e) tibble(term = NA_character_, VIF = NA_real_))
bp_test <- lmtest::bptest(m_base)
par(mfrow = c(2,2)); plot(m_base); par(mfrow = c(1,1))
# Winsorized (1–99%)
wins_cols <- intersect(present, names(df0))
df_w <- df0
for (nm in wins_cols) df_w[[paste0(nm,"_w")]] <- winsor(df0[[nm]])
```

```
df_w$yr10_salary_w <- winsor(df0$yr10_salary)
rhs_w <- c(paste0(wins_cols, "_w"), "program_type")
if ("control" %in% names(df0)) rhs_w <- c(rhs_w, "control")
m_w <- lm(reformulate(rhs_w, response = "yr10_salary_w"), data = df_w)
# Log model (complete cases only) + stepAIC

    log_vars <- unique(c("yr10_salary", present, "program_type", if ("control" %in%
names(df0))
"control"))
df_log <- df0[df0$yr10_salary > 0, , drop = FALSE]
common <- intersect(log_vars, names(df_log))
df_log <- df_log[, common, drop = FALSE]
df_log <- df_log[complete.cases(df_log), , drop = FALSE]
if ("program_type" %in% names(df_log)) df_log$program_type <- factor(df_log$program_type)
if ("control" %in% names(df_log)) df_log$control <- factor(df_log$control)
m_log <- lm(reformulate(setdiff(names(df_log), "yr10_salary"), response = "log(yr10_salary)"),
data = df_log)
m_step <- MASS::stepAIC(m_log, direction = "backward", trace = FALSE)
# ===================== RANDOM FOREST =====================
# build RF using available predictors
salary_col <- "yr10_salary"
rf_predictors <- c(present, "program_type")
if ("control" %in% names(df0)) rf_predictors <- c(rf_predictors, "control")
rf_predictors <- intersect(rf_predictors, names(df0))
rf_df <- df0[, unique(c(salary_col, rf_predictors)), drop = FALSE]
names(rf_df)[names(rf_df) == salary_col] <- "salary"
if ("program_type" %in% names(rf_df)) rf_df$program_type <- factor(rf_df$program_type)
if ("control" %in% names(rf_df)) rf_df$control <- factor(rf_df$control)
rf_df <- rf_df[complete.cases(rf_df), , drop = FALSE]
rf_df <- rf_df[, sapply(rf_df, function(x) !all(is.na(x))), drop = FALSE]
y <- rf_df[["salary"]]
x <- rf_df[, setdiff(names(rf_df), "salary"), drop = FALSE]

    rf <- randomForest(x = x, y = y, importance = TRUE, ntree = 1000)
rf_imp <- as.data.frame(importance(rf)) %>%
tibble::rownames_to_column("variable") %>%
arrange(desc(`%IncMSE`))
varImpPlot(rf, main = "Random Forest Variable Importance — Program Type Focus")
print(head(rf_imp, 10))
# ==================== REPORTING ====================
glance_tbl <- bind_rows(
```

```
glance(m_base) %>% mutate(model = "baseline_raw"),
if (!is.null(form_int)) glance(m_int) %>% mutate(model = "interact") else NULL,
glance(m_w) %>% mutate(model = "winsorized"),
glance(m_log) %>% mutate(model = "log"),
glance(m_step) %>% mutate(model = "log_stepAIC")
)
tidy_all <- bind_rows(
tidy(m_base) %>% mutate(model = "baseline_raw"),
if (!is.null(form_int)) tidy(m_int) %>% mutate(model = "interact") else NULL,
if (!is.null(form_int)) tidy_int_HC3 %>% transmute(term, estimate, std.error = se_robust,
statistic = t_robust, p.value = p_robust, model) else NULL,
tidy(m_w) %>% mutate(model = "winsorized"),
tidy(m_log) %>% mutate(model = "log"),
tidy(m_step) %>% mutate(model = "log_stepAIC")
)
cat("\n===== EVA — PROGRAM TYPE SNAPSHOT =====\n")

        cat("N (Business + Healthcare) =", nrow(df0), "\n")
cat("Present predictors:", paste(present, collapse = ", "),
if (length(absent)) paste0(" | Absent: ", paste(absent, collapse = ", ")) else "", "\n")
cat("Adj R^2 — baseline:", round(summary(m_base)$adj.r.squared, 3),
if (!is.null(form_int)) paste0(" | interact:", round(summary(m_int)$adj.r.squared, 3)) else "",
" | winsor:", round(summary(m_w)$adj.r.squared, 3),
" | log:", round(summary(m_log)$adj.r.squared, 3),
" | log_stepAIC:", round(summary(m_step)$adj.r.squared, 3), "\n")
cat("BP test p-value (heteroskedasticity):", signif(bptest(m_base)$p.value, 4), "\n")
cat("\nRF top vars by %IncMSE:\n"); print(head(rf_imp, 8))
cat("========================================\n")
# handy Viewer tabs (grab screenshots)
if (interactive()) {
View(glance_tbl, title = "Model Fit (glance)")
View(tidy_all, title = "All Coefficients (incl. HC3 for interact)")
View(rf_imp, title = "RF Variable Importance")
}
# QQ plots
qqnorm(residuals(m_base)); qqline(residuals(m_base)); title("QQ — Raw OLS Residuals")
qqnorm(residuals(m_log)); qqline(residuals(m_log)); title("QQ — Log OLS Residuals")
```

Appendix B SQL

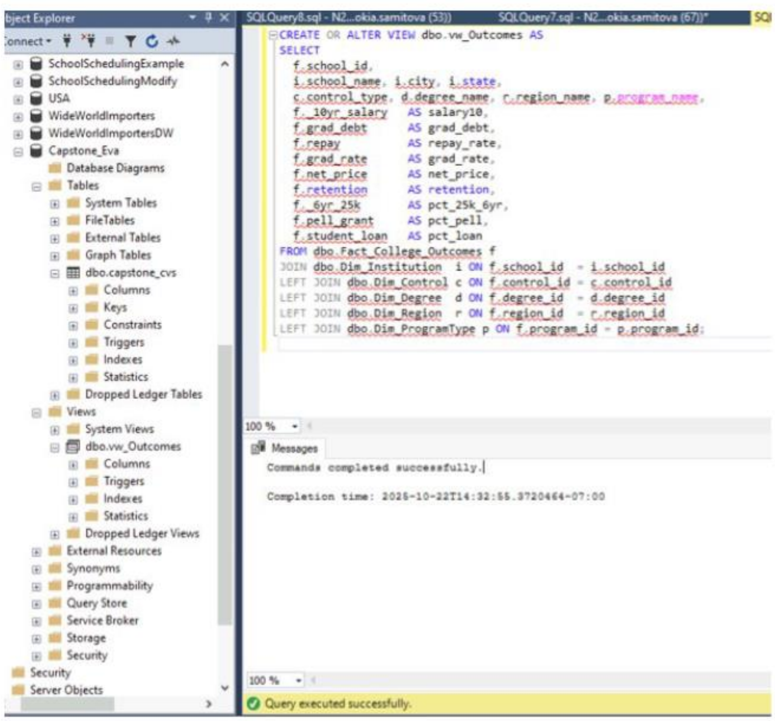
Schema Overview. Derived field for program type:

```
CASE
WHEN business/(business+health) >= 0.55 THEN 'Business'
WHEN business/(business+health) <= 0.45 THEN 'Healthcare'
ELSE 'Mixed'
END AS program_type
```

View for Tableau connection:

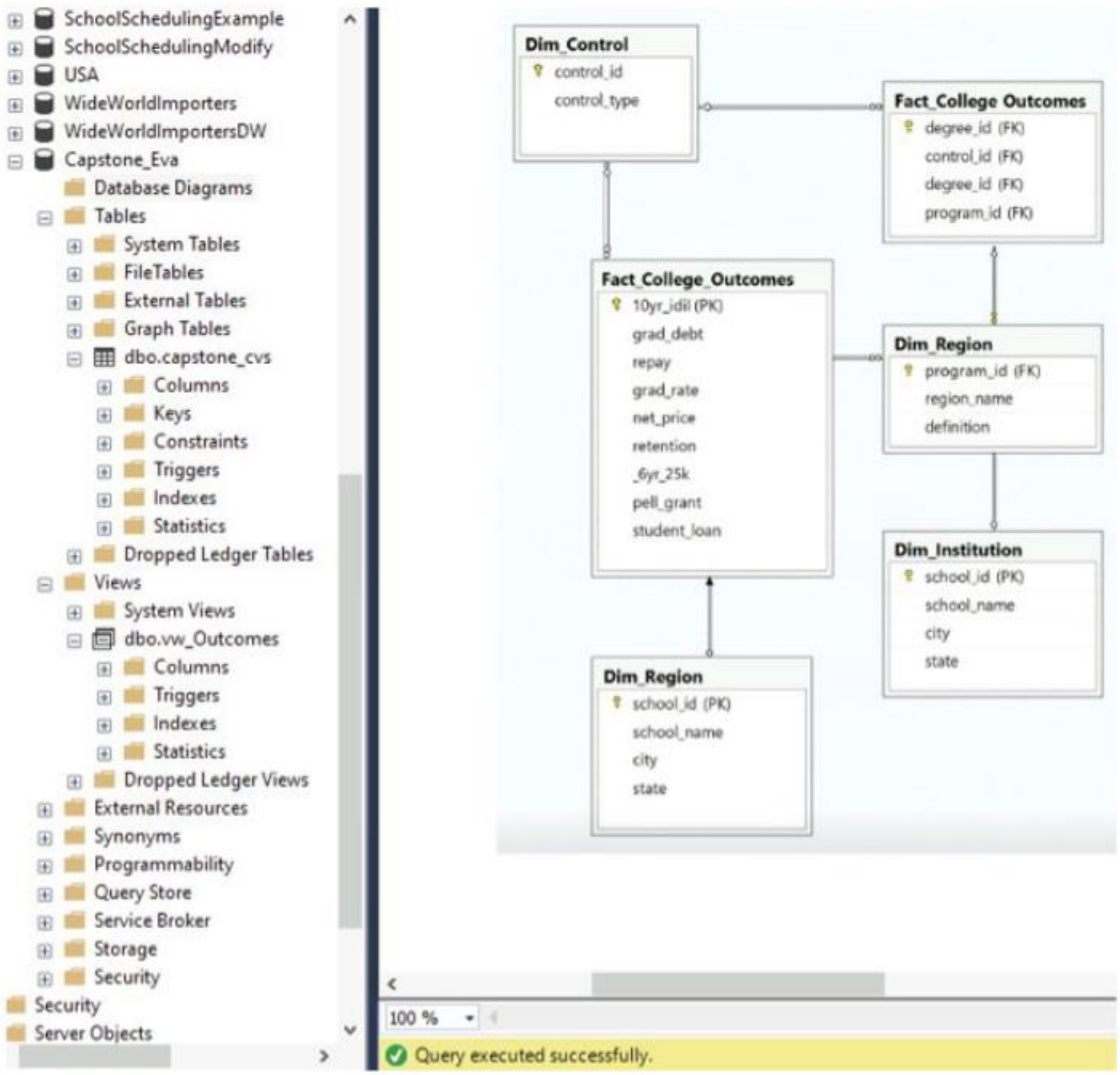
```
CREATE VIEW vw_metrics_roi AS
SELECT m.*,
CASE WHEN NULLIF(m.net_price,0) IS NULL THEN NULL
ELSE m.yr10_salary / m.net_price END AS roi,
p.program_type, i.control, i.state
FROM METRICS m
JOIN PROGRAM p ON m.program_id = p.program_id
JOIN INSTITUTION i ON p.institution_id = i.institution_id;
```

This structure enables direct Tableau connections and supports updates as new Scorecard data is added.



ER Diagram Explanation

The entity–relationship (ER) diagram below represents the Capstone_Eva database schema implemented in Microsoft SQL Server to support the data analytics and Tableau dashboard workflows. It follows a star schema structure, with Fact_College_Outcomes at the center and five supporting dimension tables providing descriptive context.



Appendix C: Tableau Visualizations

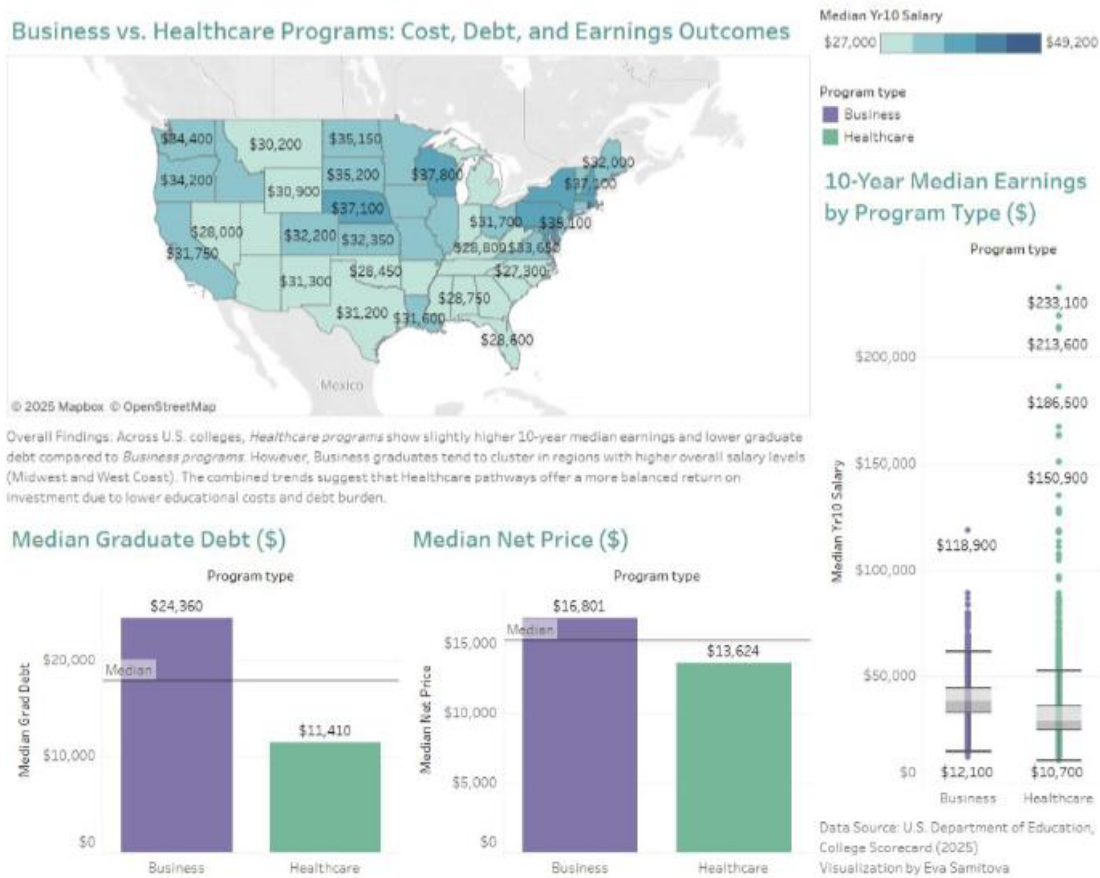


Figure C1. Business vs. Healthcare Programs — Cost, Debt, and Earnings Outcomes (Tableau Dashboard)

This dashboard compares Business and Healthcare programs across U.S. institutions, showing regional salary patterns, median debt, net price, and 10-year earnings. Business programs display higher median salaries and higher debt levels, while Healthcare programs show lower costs and more balanced ROI.

Appendix D PROJECT TIMELINE

September 24–29

- Team formation & topic selection
- Group chat created.
- Shared initial ideas and datasets.

- Team explored multiple options; agreed to choose College Scorecard dataset focusing on program-level outcomes. Early discussion on choosing an original angle (to avoid repeating previous years' topics). Downloaded dataset and shared files.

September 29 – October 2

- Project direction clarified
- Agreed the project will analyze which programs lead to stable long-term earnings.
- Discussion of required deliverables: regression, random forest, Tableau visuals, presentation.
- Instructor emphasized regression models, assumptions, normality checks, outlier detection, clear narrative
- Team saved screenshots from class to guide analysis.

October 3–10

- Data cleaning phase
- Combined multiple Scorecard files.
- Cleaned inconsistent labels for schools and programs.
- Removed missing and invalid values.
- Discussed outliers with extremely high/low earnings.
- Converted numeric variables and checked distributions.
- Validated interpretability of each variable.

October 11–20

- Feature selection and variable decisions
- Identified main predictors:
- Net Price
- Pell Grant %
- Program Tier / CIP code
- Institution Control (public/private/for-profit)
- Earnings 10–15 years after graduation
- Team clarified variable meanings:
- Institution Control = whether the school is public, private nonprofit, or private for-profit.
- Net Price = average amount students pay after accounting for grants/scholarships.
- Pell Grant share = % of low-income students receiving Pell Grants.

October 20–26

- Modeling work

- Ran multiple linear regression models.
- Checked:
- VIF (multicollinearity)
- residual plots
- normality
- heteroskedasticity
- Tested random forest to compare predictive power.
- Discussed removing or keeping outliers.
- Team shared results, screenshots, and outputs.

October 27–November 2

- Model refinement & interpretation
- Cleaned dataset finalized.
- Regression interpretation drafted.
- Final variables selected.
- Agreed on which insights will go in slides vs. written report.
- Discussed storyline: "Which programs offer the best long-term financial return?"

November 3–10

- Visualization phase
- Tableau dashboards created.
- Visuals for:
- Net price vs. earnings
- Earnings by control type
- Pell Grant % vs. earnings
- Program clusters
- Exported visuals for presentation.

November 11–18

- Slide development
- Built structure for the PPT:
- Problem statement
- Why this matters
- Data sources
- Cleaning process
- Models & results
- Tableau visuals
- Key insights
- Why stakeholders should care

- Discussed speaker notes and simplification for a management-level audience.

November 19–24

- Final preparation
- Rehearsal planning.
- Fixing formatting, fonts, and consistent design.
- Adding model interpretations in plain English.
- Finalizing team roles for presentation:
- One person on data collection/cleaning
- One on modeling
- One on visualization + conclusion
- Confirmed timing: 6–15 minutes + Q&A.

Our team collaborated consistently throughout the quarter using a dedicated WhatsApp group chat. Early messages focused on exploring multiple project ideas, comparing datasets, and reviewing previous examples. Within the first week, we agreed to use the College Scorecard dataset and define our research question: Which academic programs lead to strong long-term earnings for graduates, and what factors predict these outcomes?

Throughout the following weeks, our communication centered on data cleaning, combining multiple files, correcting inconsistent labels, removing missing values, converting numerical variables, and addressing extreme outliers. We discussed key features to include, such as institution control, net price, earnings, and Pell Grant share, and ensured that each variable was interpretable and relevant to our model.

The modeling stage included linear regression, diagnostic tests (multicollinearity, residual behavior, normality), and a random forest model for comparison. The team shared intermediate results, screenshots, concerns, and interpretations to align our narrative and ensure accuracy. We then shifted to visualization work, coordinating Tableau dashboards that highlighted relationships between earnings and institutional or financial variables.

In the final stage of communication, we focused on presentation development, creating slides, organizing insights, designing visual elements, and preparing a clear storyline suitable for a management-level audience. Responsibilities were divided among data preparation, modeling, and visualization. By the end of the chat history, the team had a complete draft of the presentation and a shared understanding of key messages, conclusions, and next steps.